PROJECT REPORT

IMPORT DATA USING TRANSFORM MAPS

ServiceNow Micro Project

Project Area	ServiceNow Data Import & Management
Platform	ServiceNow
Main Concepts	Import Sets, Transform Maps, Coalesce, Reports & Dashboards
Target Data	Employee Test Table
Report Type	Academic / Practical Project Report

Prepared as a practical project report

1. Abstract

This project demonstrates how employee information maintained in an external spreadsheet can be imported into ServiceNow in a structured and controlled way. The implementation uses an Import Set as a staging area and a Transform Map to move the source data into a custom Employee Test table. The project also includes Coalesce configuration so that repeated imports can update existing employee records instead of creating unnecessary duplicates. Finally, reports and a dashboard are created to make the imported information easier to understand and monitor.

The practical flow starts with preparing sample employee data in a spreadsheet, creating the required ServiceNow tables and fields, loading the spreadsheet, mapping source fields to target fields, and running the transformation. The result is then checked through the Employee Test table and Transform History. The later stages extend the basic import by adding duplicate-handling logic, reports, and a centralized dashboard.

2. Introduction

In many organizations, employee information is received from different sources such as spreadsheets or other external systems. Entering every record manually into ServiceNow can take time and may introduce mistakes. ServiceNow Import Sets and Transform Maps provide a practical way to bring this type of structured data into the platform.

In this project, a small employee dataset is used to understand the complete import process. Instead of treating the import as a one-time activity, the project also considers what happens when the same data is imported again. Coalesce is therefore used to identify existing records and control whether records should be updated or inserted. The project is then extended with simple reports and a dashboard for data visibility.

3. Objectives

- Prepare structured employee information in spreadsheet format.
- Create a custom Employee Test table in ServiceNow with the required fields.
- Create an Import Set table to receive the spreadsheet data.
- Configure a Transform Map between the Import Set and Employee Test table.
- Map source fields to their corresponding target fields and execute the transformation.
- Use Coalesce to avoid duplicate employee records during repeated imports.
- Verify inserted, updated, and ignored records through Transform History.
- Create reports based on department, location, and employee details.
- Combine the reports into an Employee Analytics Dashboard.

4. Tools and Technologies Used

Tool / Feature	Purpose
ServiceNow	Platform used to create tables, import data, transform records and build reports.

Import Sets	Temporary/staging area for data loaded from the external spreadsheet.
Transform Maps	Defines how source data is transferred into the target table.
Coalesce	Identifies existing records so repeated imports can update rather than duplicate them.
Reports	Presents employee data using charts and list format.
Dashboard	Provides a centralized view of the created reports.
Google Spreadsheet / Excel	Used to prepare and supply sample employee data.

5. System Design / Overall Flow

The project follows a simple data pipeline. The spreadsheet acts as the source, the Import Set table receives the raw data, and the Transform Map controls how the data is moved into the Employee Test table. After transformation, the target records are checked. Coalesce is then used to make repeated imports safer. Reports and a dashboard are built on the final employee data.

1	**Spreadsheet**	**Employee data prepared externally**
2	Import Set	Source data loaded into a staging table
3	Transform Map	Source fields mapped to target fields
4	Employee Test	Final employee records created/updated
5	Validation	Transform History and table records checked
6	Reports & Dashboard	Data summarized and displayed visually

6. Implementation

6.1 Preparing the Spreadsheet

A new spreadsheet is created with sample employee information. The prepared sheet is downloaded in Excel (.xlsx) format so that it can be used as the source file for the ServiceNow import process.

6.2 Creating the Employee Test Table

A custom table named **Employee Test** is created. Its backend name is **u_employee_test**. The form is then configured with the fields Employee ID, Employee Name, Email, Department, and Location. These fields provide the target structure for the imported employee records.

6.3 Creating the Import Set Table

Using **Load Data**, an Import Set table named **Employee Import** is created. The Import Set acts as the staging area where the spreadsheet data is first received before transformation.

6.4 Creating the Transform Map

A Transform Map named **Sample Spreadsheet Import** is configured. The Employee Import table is selected as the source and Employee Test is selected as the target. Auto Map Matching Fields and Mapping Assist are used to establish the required source-to-target field mappings. The map is then saved and used to transform the imported data.

6.5 Running the Transformation

After the mapping is completed, the Transform action is executed. A successful transformation creates records in the Employee Test table. The target table is opened afterward to verify that the employee records have been created correctly.

7. Coalesce and Duplicate Handling

One of the more important practical improvements in the project is the use of Coalesce. In a repeated import scenario, the same employee may appear again with updated information. Simply inserting every row could create duplicate employee records.

To handle this, a field in the Transform Map is configured with Coalesce set to true. In the project, the Employee ID is used as the identifying value. When the same Employee ID is received again, ServiceNow can identify the existing record and update it instead of creating another record.

During testing, four rows are uploaded in a later spreadsheet. Two represent new employee IDs and are inserted, while two existing IDs contain changed name/email information and are updated. When the same sheet is imported again without changes, the project records the rows as ignored rather than inserting duplicates. This demonstrates the intended effect of the Coalesce configuration.

8. Reports Created

The project is extended beyond data loading by creating three reports on the Employee Test table.

Report	Type	Configuration / Purpose
Employees by Department	Pie Chart	Groups employees by Department and uses Count as the aggregation.

Employees by Location	Bar Chart	Groups employees by Location and uses Count as the aggregation.
Employee List Report	List	Displays Employee ID, Employee Name, Email, Department and Location.

Each report is run and saved after its configuration is completed. The reports provide different views of the same employee data, making the information easier to interpret than a plain table alone.

9. Dashboard Creation

A dashboard named **Employee Analytics Dashboards** is created to bring the three reports together. The saved reports are added to the dashboard through the report sharing and Add to Dashboards option. Employees by Department, Employees by Location, and Employee List Report are therefore available from one centralized view.

The project material also shows that dashboard access and sharing can be controlled for users, groups, and roles. This is useful in a real ServiceNow environment because employee information may need to be visible only to authorized users.

10. Testing and Results

Testing is carried out at multiple points instead of assuming that a successful import message is enough. The imported records are checked in the Employee Test table, while Transform History is used to understand what happened to the uploaded rows.

- Initial spreadsheet records were successfully loaded through the Import Set.
- The Transform Map successfully transferred the mapped data into the Employee Test table.
- The first repeated-data test demonstrated both inserted and updated records.
- The same spreadsheet was imported again to verify that unchanged duplicate records were not inserted again.
- Transform History was used to observe inserted, updated, and ignored results.
- Reports were successfully generated for department, location, and employee-list views.
- The three reports were added to the Employee Analytics Dashboard.

11. Key Observations

The project shows that data migration is not only about moving rows from one place to another. The structure of the target table, the mapping logic, and the method used to identify existing records all affect the quality of the final data.

The Coalesce test is particularly useful because it demonstrates how the same import process can behave differently depending on whether a record is new or already present. This makes the project closer to a practical enterprise data-import scenario rather than a one-time demonstration.

The addition of reports and a dashboard also shows the next step after successful data migration: making the resulting information useful to administrators and other authorized users.

12. Conclusion

The project successfully demonstrates an end-to-end employee data import process in ServiceNow using Import Sets and Transform Maps. Starting with an Excel spreadsheet, the data is loaded into an Import Set, mapped to a custom Employee Test table, transformed, and validated. The implementation is further strengthened through Coalesce, which supports updating existing employee records and helps prevent duplicate entries during repeated imports.

The project is also extended with three reports and an Employee Analytics Dashboard. These additions provide a more useful view of the imported data by showing department-wise distribution, location-wise distribution, and the employee record list. Overall, the project provides practical exposure to ServiceNow data import, transformation, validation, reporting, and dashboard configuration.

13. Future Enhancements

- Add stronger data validation rules before transformation.
- Handle invalid or incomplete employee records through controlled error handling.
- Add additional employee fields when required by the organization.
- Introduce role-based access for reports and dashboards.
- Schedule regular imports if the source data is received periodically.
- Add more dashboard indicators for import success, errors, updates, and new records.

Appendix: Configuration Reference

Item	Value
Target Table	Employee Test
Target Table Name	u_employee_test
Import Set Table	Employee Import
Import Set Name	u_employee_import
Transform Map	Sample Spreadsheet Import
Target Fields	Employee ID, Employee Name, Email, Department, Location
Coalesce	Enabled on an identifying field (Employee ID in the project)
Dashboard	Employee Analytics Dashboards

Source basis: the uploaded project material titled “Import Data using Transform Maps (Spreadsheet)”. The report has been reorganized and rewritten into a natural project-report format while retaining the project’s stated workflow, configuration names, testing observations, reports, dashboard, and conclusion.